



University Institute of Engineering

Department of Computer Science & Engineering

EXPERIMENT: 4

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BRANCH: BE-CSE

SECTION / GROUP: KRG_3A

SEMESTER: 5TH

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SUBJECT NAME: ADBMS

1. Consider a relation R having attributes as R(ABCD), functional dependencies are given below:

AB-

>C C-

>D D-

>A

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Ans:

R (A, B, C, D)

Closure:

A+ $\square$ A

B+ $\square$ B

C+ $\square$ C, D, A

AB+ $\square$ A, B, C, D

AC+ $\square$ A, C, D

AD+ $\square$ A, D,

BC+ $\square$ B, C, D, A

BD+ $\square$ B, D, A, C

CD+ $\square$ C, D, A

Candidate Keys: AB, BC, BD

Prime Attributes: A, B, C, D

Non-prime Attributes:

Normal Form: 3NF

2. Relation R(ABCDE) having functional dependencies

as: $A \rightarrow D$

$B \rightarrow A$

$BC \rightarrow D$

$AC \rightarrow$

BE

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Ans:

R (A, B, C, D, E)

Closure:

$A^+ \sqsubset A, D$

$B^+ \sqsubset B, A, D$

$C^+ \sqsubset C$

$AB^+ \sqsubset A, B, D$

$AC^+ \sqsubset A, C, D, B, E$

$AD^+ \sqsubset A, D$

$BC^+ \sqsubset B, C, A, D, E$

Candidate Keys: AC, BC

Prime Attributes: A, B, C

Non-prime Attributes: D, E

Normal Form: 1NF

3. Consider a relation R having attributes as R(ABCDE), functional dependencies are given below:

$B \rightarrow A$

$A \rightarrow C$

$BC \rightarrow$

D

$AC \rightarrow BE$

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Ans:

R (A, B, C, D, E)

Closure:

$A^+ \sqsubset A, C, B, E, D$

$B^+ \sqsubset B, A, C, D, E$

$C^+ \sqsubset C$

$D^+ \sqsubset D$

$E^+ \sqsubset E$

Candidate Keys: A, B

Prime Attributes: A, B

Non-prime Attributes: C, D, E Normal

Form: BCNF

4. Consider a relation R having attributes as R(ABCDEF), functional dependencies are given below:

A-

$\rightarrow$ BCD

BC-

$\rightarrow$ DE B-

$\rightarrow$ D

D $\rightarrow$ A

Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Ans:

R(A, B, C, D, E, F)

Closure:

A⁺ $\sqsupseteq$ A, B, C, D, E

B⁺ $\sqsupseteq$ B, D, A, C, E

C⁺ $\sqsupseteq$ C

D⁺ $\sqsupseteq$ D, A, B, C, E

E⁺ $\sqsupseteq$ E

F⁺ $\sqsupseteq$ E

AF⁺ $\sqsupseteq$ A, B, C, D, E, F

BF⁺ $\sqsupseteq$ B, F, D, A, C, E

CF⁺ $\sqsupseteq$ C, F

DF⁺ $\sqsupseteq$ D, F, A, B, C, E

Candidate Keys: AF, BF, DF

Prime Attributes: A, B, D, F

Non-prime Attributes: C, E

Normal Form: 1NF

5. Designing a student database involves certain dependencies which are listed below:

X $\rightarrow$ Y

WZ -

$\rightarrow$ X WZ

$\rightarrow$ Y Y -

$\rightarrow$ W Y -

$\rightarrow$ X

Y $\rightarrow$ Z

The task here is to remove all the redundant FDs for efficient working of the student database management system.

Ans:

$R(W, X, Y, Z)$

Closure:

$X^+ \sqsupseteq X, Y, W, Z$

$Y^+ \sqsupseteq Y, X, W, Z$

$WZ^+ \sqsupseteq W, Z, X, Y$

Candidate Keys: X, Y, WZ

Prime Attributes: X, Y, W, Z

Non-prime Attributes:

Normal Form: BCNF

6. Debix Pvt Ltd needs to maintain database having dependent attributes ABCDEF. These attributes are functionally dependent on each other for which functional dependency set F given as:

$A \rightarrow BC$

$D \rightarrow E$

$BC \rightarrow D$

$A \rightarrow D$

Consider a universal relation $R_1(A, B, C, D, E, F)$ with functional dependency set F, also all attributes are simple and take atomic values only. Find the highest normal form along with the candidate keys with prime and non-prime attribute.

Ans:

$R(A, B, C, D, E, F)$

Closure:

$A^+ \sqsupseteq A, B, C, D, E$

$B^+ \sqsupseteq B$

$C^+ \sqsupseteq C$

$D^+ \sqsupseteq D, E$

$AF^+ \sqsupseteq A, B, C, D, E, F$

Candidate Keys: AF

Prime Attributes: A, F

Non-prime Attributes: B, C, D, E

Normal Form: 1NF